An Algorithm for Automated Feature Detection of Eddy Current Defects Using Sum of Gaussian Method

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***Abstract*— This article describes an algorithm used to automatically detect differential coil eddy current (EC) defect signal features using a sum of Gaussians approach. Defect signals are decomposed into positive and negative portions then a thresholded binary image is constructed using the Parzen–Rosenblatt window technique. Mathematical morphology operators are then used to denoise the thresholded binary image. Peaks are detected in the denoised binary images using a simplified variant of the K-means algorithm. The resulting parameters obtained are input into a constrained optimization algorithm to obtain an estimate of the denoised EC defect signal using a minimum summed-square error objective function. The algorithm performance is investigated over a range of electrostatically discharged machine (EDM) notch return signal windows that vary in coil diameter, test material, test frequency, and number of defects in the investigated region.**

***Index Terms*— Algorithms, eddy current testing, Gaussian mixture model, level set, morphological operations, optimization.**

I. Introduction

EDDY current inspection (ECI) is one nondestructive inspection (NDI) method commonly used to investigate various conditions of metallic components [1], [2]. The basic process entails injecting a known electromagnetic signal into the component (called the unit under test—UUT) and capturing the altered return signal for analysis. There are numerous variants of the ECI method in terms of drive sensor/signals used, receive sensors employed, and detection and characterization algorithms. An extremely important property of ECI systems is the ability to detect features of impedance anomalies (defect signatures) in the return signal from the UUT and to characterize those signatures in terms of common properties like length, width, and depth of the defect. In many instances, a singular feature of the defect signature (e.g., peak amplitude) is used in the defect characterization process.

The anomaly detection algorithm described in this article pertains to the return signals typically seen in an ECI method that uses reflection differential coil (D-coil) inspection probes which are commonly used in general ECI applications. The typical D-coil ECI probe design is comprised of two split receive coils wound 180° out of phase with respect to each other and encircled by a drive coil. A picture of the arrangement is shown in Fig. 1.

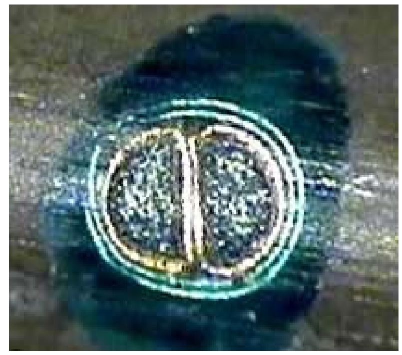

Fig. 1. D-coil eddy current probe [3]. The green coil is the probe drive coil and inside this are the two receive coils.

An example of the automated D-coil ECI technique employs an electromagnetic drive signal of known amplitude and frequency, with the probe perpendicularly oriented and placed in close proximity to the UUT to minimize the effects of lift-off which can attenuate the return signal. The receive sensor coils collect the modulated return signal which is either a direct or representative measure of the UUT impedance variation in the reception field of the coils.

Using the aforementioned approach, typically the ECI probe is robotically scanned over the UUT at known speed and the resulting received measurements are quadrature (IQ) demodulated and digitized for offline analysis. Material defects in the UUT will generate a signal similar to the one indicated in Fig. 2 (vertical component).

One immediate observation of the defect signature in Fig. 2 is the appearance of Gaussian-like peaks that comprise the signal. Defect detection would be a fairly simple process if all ECI receive signals presented defect responses like the example in Fig. 2. However, the ECI technique has numerous factors that can affect the defect signal appearance like separation of the probe from the UUT (lift-off), coil orientation relative to the surface [4], receive coils geometrical relationship relative to the drive coil, defect orientation related to the receive coils, scan speed, and the volume of the UUT defect that is present in the reception field of the ECI probe.

To illustrate this last point, a series of plots are presented in Fig. 3 which indicate responses typically seen for the vertical

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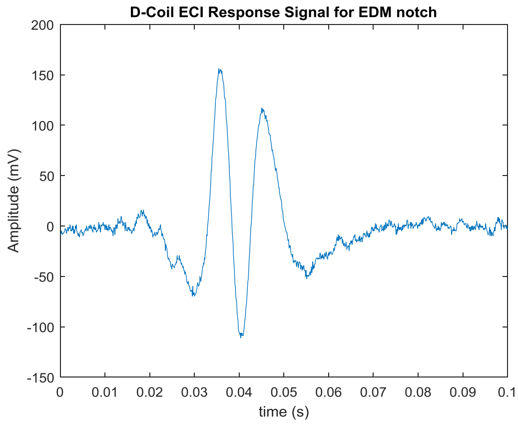

Fig. 2. D-coil ECI return signal scanning over EDM notch.

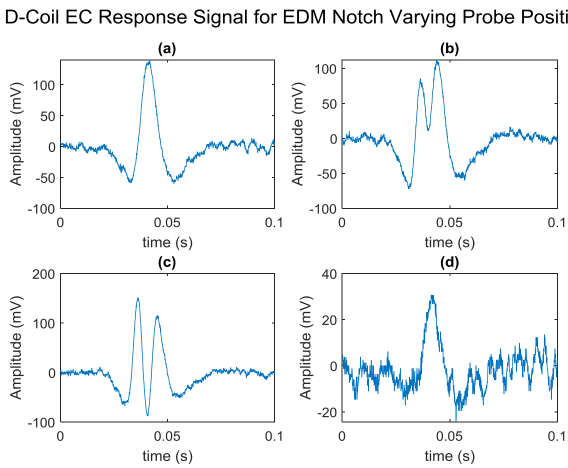

Fig. 3. D-coil ECI response varying probe position. (a) Probe is approaching defect region as sensed by leading receive coil, (b), (c) both receive coils sensing defect, (d) probe is leaving the defect region. (a)–(d) Note the changes in defect response shape and amplitude which are primarily due to the volume of the defect in the reception fields of the coils.

component of a D-coil-type probe receive signal when the probe is indexed across a small (relative to the coil diameter) EDM notch while keeping the drive signal amplitude and frequency fixed.

As indicted in Fig. 3, changing the defect volume in the region of the receive coil fields results in significant changes in the defect signature shape.

In addition to the parameters that can affect the defect signature appearance, there are nonfunctional parameters that come into play such as constraints on inspection time that make defect detection schemes that achieve increased detection rate through multiple looks at the target somewhat intractable for automated inspection scenarios.

Due to the fact that there are many factors that affect a given ECI defect signal as well as nonfunctional requirements like inspection throughput, a robust defect detection scheme would be useful in not only reliably detecting attributes of the defect signal, but the algorithm should be able to describe the defect features in some useful manner to provide characterization algorithms greater information to determine better pass/fail decisions thereby reducing the false call rate. In addition, such an algorithm should have the ability to obtain said information without requiring additional static looks at the region under investigation when nonfunctional requirements like inspection throughput are of concern.

To this end, this study describes a defect feature detection technique that transforms a given D-coil ECI return signal window using the Parzen–Rosenblatt [5], [6] approach along with the concept of level sets [7] to create a binary image. Next, binary mathematical morphology (MM) operations are used to denoise the signal and isolate possible areas of interest. A variant of the K-means algorithm is then used to determine the approximate location of candidate Gaussian-like peaks in the denoised signal. Last, the candidate defect signature peak center locations are used to model the defect signature using the sum of Gaussians (SoGs) method.

A literature review was conducted focusing on the use of MM filtering and image thresholding techniques of ECI defect signals. In addition, a search was conducted for prior approaches to fitting signals to Gaussian functions. The results of this search follow.

A. Literature Review

1) Defect Detection Using MM Methods: Nonbinary MM methods have been used for defect signature detection in the presence of edge geometries of jet engine components [8] and in the analysis of C-scan images of outer surfaces of aircraft wheels [9]. MM filtering techniques have also been used in the analysis of cracks in welded structure using pulsed EC coupled with thermal imaging [11]. Last, MM filtering operations have been used in the study of cracks in pipe structures utilizing image analysis on the ECI signal impedance plane representation [12].

2) Gaussian Curve Fitting Methods: Numerous Gaussian curve fitting approaches utilize the fact that the exponential term of the Gaussian function is quadratic. Using this feature, the curve fitting problem is reduced to a quadratic line fit by taking the logarithm of both sides of the Gaussian equation [13], [14]. Implicit in the efficacy of these approaches is that the region of interest is sufficiently isolated and not influenced strongly by other Gaussian signals in the vicinity and that sufficient data points exist in the region of interest to reliably determine a candidate fit.

Other approaches to curve fitting utilize an SoG approach with one notable example being a technique that transforms the original signal into a scale-space representation [15] and then processes the transformed signal to obtain the initial Gaussian parameters which are then optimized via the Marquardt algorithm [16]. The SoG approach has also been used to characterize EC defects in terms of position, width, and depth using collections of Gaussian pairs and an artificial neural network [17].

The presented algorithm is relevant for a number of reasons.

1) Of the MM filtering-based ECI filtering techniques surveyed, an interesting trend was observed: nonbinary MM approaches were used when investigating an individual ECI scan whereas binary MM approaches were used when filtering C-scan (multiple single scans) ECI

images. However, in both cases the filtering kernel used was based on a model of the entire ECI defect signature. Careful inspection of the defect signal presented in Figs. 1 and 3 indicate that a D-coil return signal could be modeled using an SoG approach. The presented algorithm utilizes only an MM filtering kernel modeled on a hypothesized portion of the EC defect signature as a means to build a complete receive defect signal model.

2) The number of data points that represent an EC defect signature can be rather small, corrupted by colored noise, and in the immediate vicinity of other Gaussian-like signals of interest. This can present problems for algorithms that utilize quadratic line fitting. In addition, due to probe geometry and scan speed the defect signature width (for small defects) typically has only a small range of σ values of typical interest. Scale-space filtering approaches could possibly yield SoG-fitted parameters that are outside the realm of interest.
3) Modeling the EC defect signal using Gaussian pairs may be acceptable for the probe used in [17]; however, D-coil-type receive signals are often asymmetric. This study makes no assumption as to the number of Gaussian-like peaks in the EC defect signature.
4) As stated previously, in many applications an EC defect signature feature like amplitude [18] is used to characterize the defect. The proposed algorithm provides much more information about the defect signature that can greatly improve the reliability and precision of the characterization processes.

The remainder of this report is organized as follows. The data sources investigated and the proposed algorithm are detailed in Section II. The investigation results and discussion are provided in Section III. Finally, Section IV provides conclusions that were drawn from the study as well as ideas for future research.

II. Methodology

A. Data Sources

The data sets used for this study were collected using the USAF Eddy Current Inspection System (ECIS) [19] during various probe calibration operations on differing materials and equipment configurations.

During the probe calibration operation, the probe is robotically indexed in known increments across a metallic surface containing EDM notches of known size. Table I provides a summary of the ECI parameters used for collection of the data sets and Fig. 4 provides a simplified diagram of the inspection setup.

DS1 consists of data from three separate calibration sequences in which the volume of the EDM defect in the region of the probe reception field varies. In some cases, the drive signal amplitude varies. The result is a set that provides varying morphologies of the return signal. DS1 was collected with a test frequency of 2 MHz (which is common to the ECIS) at a scan rate of 5 in/s and the window under investigation contained one EDM notch (dim.: 0.010 (l) × 0.004 (w) × 0.006(d) in) in all instances.

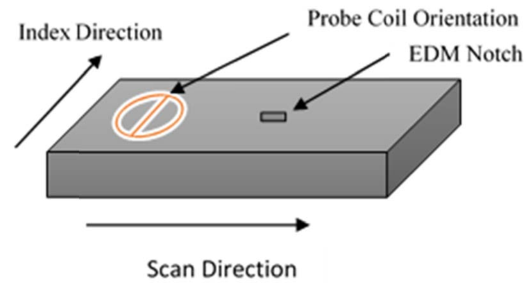

Fig. 4. Simplified diagram of ECI setup. The centerline of the D-coil is transverse to the EDM notch on the UUT.

TABLE I
Study Data Set Summary

Dataset ID	Nominal Coil Diameter (In.)	Material	Coil/Notch Orientation	No. Samples
DS1	0.080	Inconel	Transverse	50
DS2	0.040	Titanium	Transverse	50

DS2 consists of data from seven separate calibration sequences in a manner similar to that described for DS1. However, DS2 was collected with a test frequency of 6 MHz (due to the effect of increased surface noise of titanium at lower frequencies) at a scan rate of 0.83 in/s and the window under investigation contained two EDM notches (dim.:0.010 (l) × 0.004 (w) × 0.006(d), dim.: 0.010 (l) × 0.004 (w) × 0.009(d) in) in all cases.

B. Feature Detection Algorithm

A diagram of the feature detection algorithm is provided in Fig. 5 with a detailed description of the algorithm stages following.

1) Gaussian Image Generation and Binarization: Table II provides pseudocode for the Gaussian image generation and binarization process.

As indicated in Table II, the original signal is windowed and partitioned into positive and negatives values with each partitioned signal normalized onto the interval [0,1].

Next, for each partitioned signal, a 2-D signal image (dimension $n \times n$) is created by using the spatial/temporal location in the window and amplitude of each signal data point as the mean and amplitude of a Gaussian function of known standard deviation (σ_e). The fitted functions are arranged in columnar format. With this arrangement, the original signal values are located along the main diagonal of the constructed signal image.

Lastly, a thresholding operation is performed on the normalized Gaussian signal image such that values above a prespecified threshold (α) referenced to the maximum amplitude value in the window, are set to unity, and values below the threshold value set to zero. This creates a binary image which

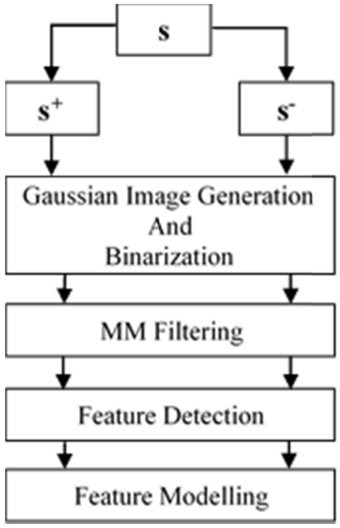

Fig. 5. Diagram of algorithm operations.

can be viewed as indicative of a top-down view of the original signal.

a) Selection of σ_e: The value of σ_e is selected to be consistent with the desired defect feature of interest. For defects that are small compared to the coil diameter σ_e can be calculated as

Let d_c = probe coil diameter and υ_s = scan rate

$$\sigma_e = \frac{0.5 d_c}{\upsilon_s}$$

b) Selection of α: The value of α is selected to determine the neighborhood size around a detected peak. If the value is too small, unwanted noise will be included in the binary Gaussian image. If the value is too large, useful portions of the detect peak neighborhood are excluded from analysis.

Using the fact that the data sets in the study have defects that occur in known regions (Figs. 6 and 9) we define a matrix $\mathbf{S}$ for a data set $\mathbf{D}$ such that each row i of $\mathbf{S}$ is

$$[\mathbf{S}]_i = \mathbf{s}_i^+ + \mathbf{s}_i^-, \quad \forall \mathbf{s}_i \in \mathbf{D}.$$

Next, submatrices are extracted such that

$$\mathbf{S}_D = [\mathbf{S}]_{\text{defect\_region}}, \quad \mathbf{S}_N = [\mathbf{S}]_{\text{noise\_region}}$$

Defining the operator

$\bar{S}$ as the mean of the elements of the matrix $\mathbf{S}$, then

$$\alpha = \bar{S_N} / \bar{S_D}.$$

2) MM Filtering: Once the binary signal images are created, the images are subjected to the binary erosion and dilation MM operators [20] using a structural element that consists of a Gaussian signal image of smaller width than the original signal. The pseudocode for the filtering operation is provided in Table III.

The MM operations have the result of removing/reducing non-Gaussian-like peaks from the signal window under study. An example of the Gaussian image generation, binarization, and MM filtering operations are provided in Figs. 6 through 8.

TABLE II
GAUSSIAN IMAGE GENERATION AND BINARIZATION PSEUDOCODE

Given: *signal vector*: $\mathbf{s} = [s_0, s_1, \ldots, s_{n-1}]^T$ *sampling instant vector*: $\mathbf{t_s} = [t_s = 0, t_1, \ldots, t_{n-1}]^T$ *Excpected Peak Width*: σ_e *Binary Image Threshold*: α	
Step	**Description**
1	Partition s such that $\mathbf{s} = \mathbf{s}^+ - \mathbf{s}^-$ Where $\mathbf{s}[\mathrm{k}]^+ = \begin{cases} \mathbf{s}[\mathrm{k}], & \mathbf{s}[\mathrm{k}] > 0 \\ 0, & otherwise \end{cases}$ $\mathbf{s}[\mathrm{k}]^- = \begin{cases} -\mathbf{s}[\mathrm{k}], & \mathbf{s}[\mathrm{k}] < 0 \\ 0, & otherwise \end{cases}$ $0 \le k \le n-1, k \in \mathbb{Z}$
2	Find the maximum values in each partitioned signal: $\mathbf{s}_{max}^+ = \max(\mathbf{s}^+), \quad \mathbf{s}_{max}^- = \max(\mathbf{s}^-)$
3	Normalize the partitioned signals $\mathbf{s}_n^+ = \frac{\mathbf{s}^+}{\mathbf{s}_{max}^+}, \quad \mathbf{s}_n^- = \frac{\mathbf{s}^-}{\mathbf{s}_{max}^-}$
4	For each partitioned signal vector form a Gaussian Image Matrix (e.g. for positive part) $\mathbf{g}_i^+ = \mathbf{s}_{n,i}^+ \exp\left(-\frac{(\mathbf{t_s} - \mathbf{t}_{s,i})^2}{\sigma_e^2}\right),$ $for\ 0 \le i \le n-1$ $\mathbf{G}^+ = [\mathbf{g}_0^+ \ \ldots \ \mathbf{g}_{n-1}^+]$
5	Create the partitioned vector Binary Image Matrices such that $\mathbf{B}^+(i,j) = \begin{cases} 1, & \mathbf{G}^+(i,j) \ge \alpha \\ 0 & otheriwise \end{cases}$
	$\mathbf{B}^-(i,j) = \begin{cases} 1, & \mathbf{G}^-(i,j) \ge \alpha \\ 0 & otheriwise \end{cases}$

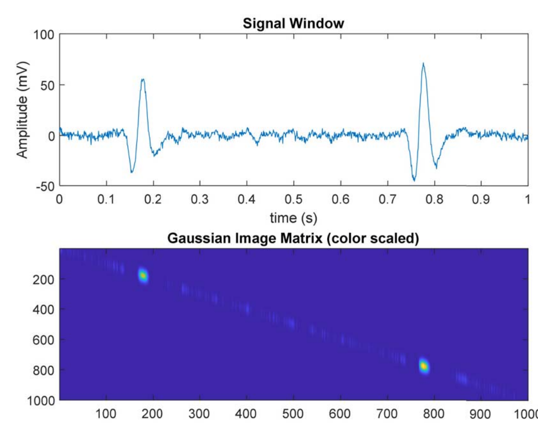

Fig. 6. Example Gaussian image matrix (scaled color). The top plot and bottom image show two EDM defect responses (0.1 s $\le t \le$ 0.3 s, 0.7 s $\le t \le$ 0.9 s) of a signal from DS2. Gaussian image is of s^+ partitioned exemplar signal.

3) Feature Detection: After the binary signal images are processed by the MM operations, a simplified variant of the K-means algorithm [21] is applied to the resulting images

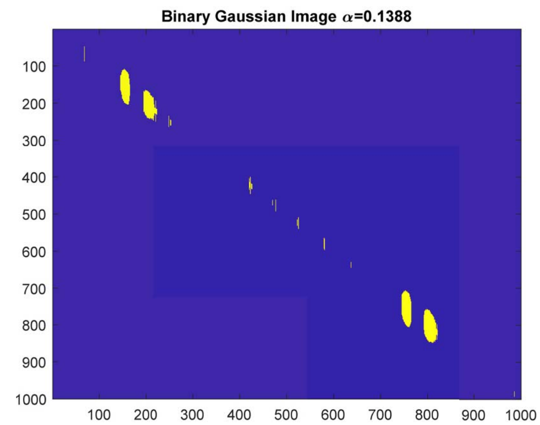

Fig. 7. Binary Gaussian image of exemplar signal. The noise content is characterized as vertical lines (low spatial relationship to adjacent samples) and patterns larger/smaller than the desired defect feature (possibly surface noise).

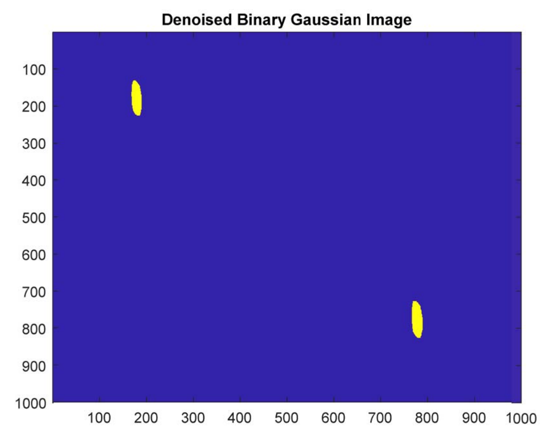

Fig. 8. Filtered exemplar signal. The noise sources in Fig. 8 are removed leaving two candidate positive peaks from s^+.

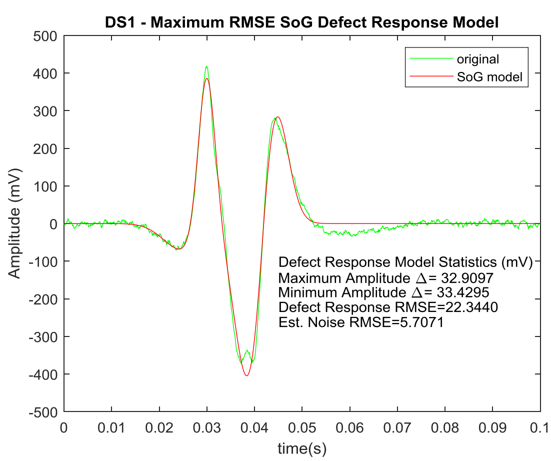

Fig. 9. DS1 maximum defect response RMSE case. There is one defect response ($0.01\ \mathrm{s} \leq t \leq 0.06$ s) present in all cases in DS1. Noise was estimated over the interval ($0.07\ \mathrm{s} \leq t \leq 0.1$ s). The algorithm detected four defect features. However, there are possible discrepancies in the response maximum and minimum peaks that simple peak analysis algorithms would not detect.

to determine the location and number of the initial Gaussian peak centers. Table IV provides pseudocode for the initial peak determination portion of the algorithm.

The initial peak centers are used as an initial guess for the Gaussian μ parameter for a given peak and the region

TABLE III
MM FILTERING ALGORITHM

Given:	
Binary Gaussian Image Matrix: $\mathbf{B}_{\mathrm{nxn}}$ *Excpected Peak Width*: σ_e *Structuring Element Width Delta*: Δ *SE Binary Image Threshold*: β	
Step	**Description**
1	Define Structuring Element (SE) width $\sigma_{SE} = \sigma_e - \Delta$
2	Form SE matrix from known Gaussian signal
	$\mathbf{f}_i = \exp\left(-\frac{\left(\mathbf{t_s} - \mathbf{t}_{s,\frac{n-1}{2}}\right)^2}{\sigma_{SE}^2}\right),$ $for\ 0 \leq i \leq n-1$ $\mathbf{F} = [\mathbf{f}_0 \quad \dots \quad \mathbf{f}_{n-1}]$
3	Create the associated Binary SE matrix and obtain submatrix containing the SE as the minimum bounding rectangle (MBR). $\mathbf{B}_{SE}(i,j) = \begin{cases} 1, & \mathbf{F}(i,j) \geq \beta \\ 0 & otheriwise \end{cases}$ $\mathbf{SE} = MBR(\mathbf{B}_{SE})$
4	Perform the MM open (erode and dilation) operation on the signal Binary Gaussian Image Matrices: $\mathbf{B}_d^+ = \mathbf{B}_{SE} \circ \mathbf{B}^+ = (\mathbf{B}_{SE} \ominus \mathbf{B}^+) \oplus \mathbf{B}^+$ $\mathbf{B}_d^- = \mathbf{B}_{SE} \circ \mathbf{B}^- = (\mathbf{B}_{SE} \ominus \mathbf{B}^-) \oplus \mathbf{B}^-$

boundary vector $\mathbf{i}$ as identified in Table IV was used to determine the width of the cluster which is representative of peak width or standard deviation parameter σ.

By partitioning the original signal into positive and negative portions, adequate separation in the point clusters is obtained that greatly reduces cluster misclassification (and consequent inaccurate peak center calculations).

The results of application of the K-means algorithm are a set of cluster centers (spatial or temporal locations) and widths that along with the associated amplitude values will comprise the initial means, standard deviations, and amplitudes for the signal fitting portion of the algorithm.

4) Feature Modeling: After obtaining the initial Gaussian parameters from the binary Gaussian images, the nonlinear constrained optimization [22] algorithm *"fmincon"* in MATLAB was used in a two-phase process to determine the estimated optimal model of the parameters to the original signal.

During the first phase, each Gaussian parameter set was independently modeled to the portion of the original partitioned signal using the width information obtained by the initial Gaussian parameter determination algorithm. The second phase used the modified Gaussian parameters from the first phase and modeled the entire Gaussian parameter set to the original signal. Both optimization phases used the

TABLE IV

INITIAL PEAK DETERMINATION ALGORITHM

Step	Description
Given:	$Binary\ Gaussian\ Image\ Matrix(filtered)$: $\mathbf{B}_{\mathrm{nxn}}$
1	Find the indices of the columns in **B** where change of state occurs $\mathbf{b} = diag(\mathbf{B})$ $\mathbf{d} = abs(diff(\mathbf{b}))$ $\mathbf{i} = find\_indices(\mathbf{d} == 1)$ $where\ k = count\ of\ indicies\ in\ \mathbf{i}$ $i_0 = 1, i_{k+1} = n$ $\mathbf{i}\ is\ a\ vector\ of\ dimension\ (k+2,1)$
2	For each of the possible k+*1* cluster regions in the image **B**, find the cluster center. Let c represent the number of column indices in the j$^{\text{th}}$ cluster. Define the candidate cluster centers as $\mathbf{p}_l = (l, l),\ where\ i_{j-1} \leq l < i_j, and\ l \in \mathbb{Z}$ let $[\mathbf{r}_j, \mathbf{c}_j]_{m,2} = find(\mathbf{B}(:, i_{j-1}: i_j) == 1)$ Represent the indices of the active pixels in cluster j. Calculate the sum of the Euclidean distances for each active pixel to each candidate cluster center $\mathbf{d}_l = \sum_{k=0}^{m-1} \sqrt{(r_k - l)^2 + (c_k - l)^2}$
3	Obtain the initial cluster center for the j$^{\text{th}}$ region as the index of the minimum of the summed distances vector **d** in step 2.

summed squared error (SSE) metric as the objective function. The second phase optimization can be stated mathematically as follows.

Assuming the algorithm detected p peaks

Given

$$\text{Peak means}: \boldsymbol{\mu} = [\mu_0, \mu_1, \ldots, \mu_{p-1}]^T$$

$$\text{Peak widths}: \boldsymbol{\sigma} = [\sigma_0, \sigma_1, \ldots, \sigma_{p-1}]^T$$

$$\text{Peak amplitudes}: \mathbf{a} = [a_0 = s(\mu_0), a_1, \ldots, a_{p-1}]^T$$

$$\text{Orignal signal vector}: \mathbf{s}$$

$$\text{Sampling instant vector}: \mathbf{t_s} = [t_s = 0, t_1, \ldots, t_{n-1}]^T$$

Sum of Gaussian

$$\mathbf{s}_{\text{est}} = \sum_{i=0}^{p-1} a_i \exp\left(-\frac{(\mathbf{t_s} - \mu_i)^2}{\sigma_i^2}\right) \tag{1}$$

The objective function used to optimize each of the Gaussian mixtures is then

$$\text{SSE} = \sum_{i=0}^{n-1} (\mathbf{s} - \mathbf{s}_{est})^2, \quad \text{for } 0 \leq i \leq n-1 \tag{2}$$

TABLE V

ANALYSIS METRICS. EACH ANALYSIS METRIC IS TAKEN OVER THE SUBINTERVALS $\mathbf{s}_{\text{defect}}$, $\mathbf{s}_{\text{noise}}$ (SEE SECTION II-B1B) FROM THE ORIGINAL SIGNAL AND THE SOG MODEL (m)

Metric	Formula
Maximum Amplitude Δ	$max(\mathbf{s}_{\text{defect}}) - max(\mathbf{m}_{\text{defect}})$
Minimum Amplitude Δ	$min(\mathbf{s}_{\text{defect}}) - min(\mathbf{m}_{\text{defect}})$
Defect Response RMSE (*l* samples)	$\sqrt{\frac{\sum_{i=0}^{l-1}(\mathbf{s}_{defect} - \mathbf{m}_{defect})^2}{l}}$
Estimated Noise RMSE (*m* samples)	$\sqrt{\frac{\sum_{i=0}^{m-1}(\mathbf{s}_{noise})^2}{m}}$

TABLE VI

ALGORITHM PARAMETERS USED FOR DATA SET ANALYSIS

Parameter	DS1 Value	DS2 Value
α	0.0672	0.1388
σ_e	0.008	0.024
Δ	0.004	0.012
β	0.9	0.9

C. Analysis Metrics

One aspect of this study is to demonstrate the efficacy of the algorithm to produce SoG models that closely model the Gaussian-like aspects of the original defect response. Another aspect is to demonstrate the ability of the SoG model to provide monitoring and defect response quality of information metrics for defect detection schemes that employ only defect response amplitude analysis for characterization purposes. To this end, the analysis metrics in Table V were used.

The aforementioned algorithm was implemented in MATLAB and the data sets identified in Table I were investigated. The results and discussion follow.

III. RESULTS AND DISCUSSION

The data sets identified in Table I were investigated using the algorithm parameter values listed in Table VI.

A. DS1 Results

The maximum and minimum case defect response RMSE results for DS1 and analysis metrics are provided in Figs. 9 and 10.

B. DS2 Results

The maximum and minimum case defect response RMSE results for DS2 and analysis metrics are provided in Figs. 11 and 12.

C. Discussion

In Fig. 9, the large maximum and minimum amplitude Δ's as well as high defect response RMSE compared to the

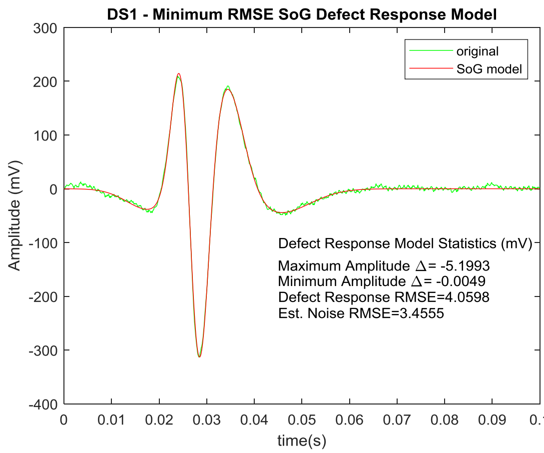

Fig. 10. DS1 minimum defect response RMSE case. The algorithm detected all five features in the defect response RMSE differs from estimated noise RMSE by 0.6048 mV.

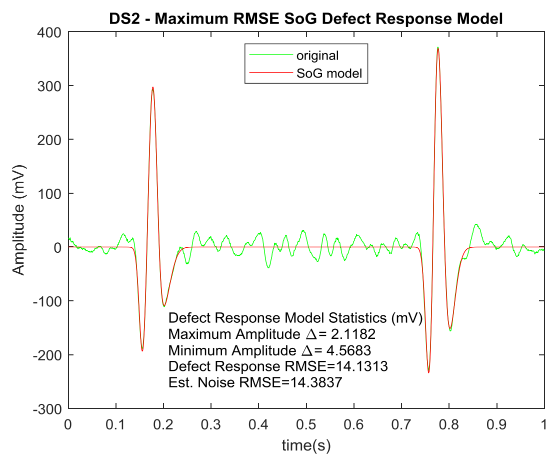

Fig. 11. DS2 maximum defect response RMSE case. The algorithm detected the defect features for both defect responses and ignored the noise present in the signal window. Analysis statistics are for larger response (0.7 s $\leq t \leq$ 0.9 s). Noise was estimated over the interval (0.3 s $\leq t <$ 0.7 s).

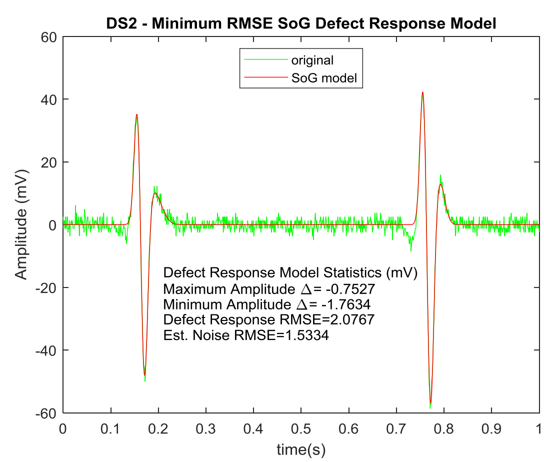

Fig. 12. DS2 minimum defect response RMSE case. The algorithm detected all defect response Gaussian-like features.

estimated noise uncover a possible quality issue with the defect response in the data set record. By contrast, Fig. 10 indicates closer agreement of the defect SoG model with the original response as the defect response RMSE is only slightly above the estimated noise by 0.6048 mV. The pattern of Fig. 10 is also witnessed in Figs. 11 and 12 again indicating close agreement of the defect SoG model to the original response.

IV. Conclusion

For ECI systems that employ defect response amplitude-based characterization designs, SoG models can provide the defect response amplitude information through the amplitude parameters as well as the peak proximities through the μ parameters. Holistically, the SoG parameters provide a defect response model that provides greater insight into the defect peak behavior that can improve ECI system robustness and may be of benefit in other ECI systems that generate defect responses with Gaussian-like features [23]–[27]. In addition, EC defect SoG models can be used to train deep learning networks to increase defect detection accuracy and possibly uncover other unforeseen defect response features. This will be a topic of future research.

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